

Coordinate Systems and Tensor Operations

Part 2 of the OT Science Formula Completion Workbook

OT Science Notes

2026-06-23

Contents

1	Coordinate Systems and Tensor Operations	2
1.1	Rectangular, Cylindrical and Spherical Coordinates	2
1.2	Basis Vectors and Scale Factors	2
1.3	Volume and Area Elements	3
1.4	Coordinate Basis, Physical Components and Tensor Components	3
1.5	Gradient, Divergence and Curl in Orthogonal Coordinates	4
1.6	Cylindrical Differential Operators	4
1.7	Spherical Differential Operators	4

1 Coordinate Systems and Tensor Operations

Coordinate systems are not just different labels for the same point. They also change how basis vectors, scale factors, area elements, volume elements and differential operators are written. Tensor operations help us keep the geometry clear when the coordinates become less simple than rectangular coordinates.

1.1 Rectangular, Cylindrical and Spherical Coordinates

Rectangular coordinates are the easiest because the basis vectors are fixed. Cylindrical and spherical coordinates are more natural for circular, axial, radial and angular symmetry, but their basis vectors change with position.

Coordinate Transformations

$$\text{Cartesian:} \quad (x, y, z).$$

$$\text{Cylindrical:} \quad x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z.$$

$$\rho = \sqrt{x^2 + y^2}, \quad \phi = \text{atan2}(y, x), \quad z = z.$$

$$\text{Spherical:} \quad x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \cos^{-1} \left(\frac{z}{r} \right), \quad \phi = \text{atan2}(y, x).$$

$$\rho = r \sin \theta, \quad z = r \cos \theta, \quad r = \sqrt{\rho^2 + z^2}.$$

1.2 Basis Vectors and Scale Factors

A coordinate basis is obtained by differentiating the position vector with respect to the coordinates. In orthogonal coordinates, the scale factors tell us how much physical length corresponds to a small coordinate change.

Basis Vectors, Scale Factors and Metric Matrices

For orthogonal coordinates (q^1, q^2, q^3) ,

$$\mathbf{g}_i = \frac{\partial \mathbf{r}}{\partial q^i}, \quad h_i = \|\mathbf{g}_i\|, \quad \hat{\mathbf{e}}_i = \frac{\mathbf{g}_i}{h_i}.$$

$$ds^2 = h_1^2 (dq^1)^2 + h_2^2 (dq^2)^2 + h_3^2 (dq^3)^2.$$

$$[g_{ij}] = \begin{pmatrix} h_1^2 & 0 & 0 \\ 0 & h_2^2 & 0 \\ 0 & 0 & h_3^2 \end{pmatrix}, \quad [g^{ij}] = \begin{pmatrix} 1/h_1^2 & 0 & 0 \\ 0 & 1/h_2^2 & 0 \\ 0 & 0 & 1/h_3^2 \end{pmatrix}.$$

$$\text{Cartesian:} \quad h_x = h_y = h_z = 1, \quad ds^2 = dx^2 + dy^2 + dz^2.$$

$$\text{Cylindrical:} \quad h_\rho = 1, \quad h_\phi = \rho, \quad h_z = 1, \quad ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2.$$

$$\text{Spherical:} \quad h_r = 1, \quad h_\theta = r, \quad h_\phi = r \sin \theta, \\ ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

1.3 Volume and Area Elements

The volume element is controlled by the product of the scale factors. This is why cylindrical and spherical integrals contain the extra factors ρ and $r^2 \sin \theta$.

Differential Elements

$$dV = h_1 h_2 h_3 dq^1 dq^2 dq^3.$$

$$dV_{\text{cart}} = dx dy dz.$$

$$dV_{\text{cyl}} = \rho d\rho d\phi dz.$$

$$dV_{\text{sph}} = r^2 \sin \theta dr d\theta d\phi.$$

For orthogonal coordinates, the coordinate surface elements are

$$dS_1 = h_2 h_3 dq^2 dq^3, \quad dS_2 = h_3 h_1 dq^3 dq^1, \quad dS_3 = h_1 h_2 dq^1 dq^2.$$

1.4 Coordinate Basis, Physical Components and Tensor Components

A vector can be represented in coordinate basis vectors or in unit basis vectors. In many engineering texts, cylindrical and spherical vector fields are written using unit basis vectors, so the listed components are physical components. In tensor notation, however, components may be covariant or contravariant depending on the basis used.

Basic Tensor Operations

For a second-order tensor A_{ij} ,

$$(\mathbf{Ax})_i = A_{ij}x_j, \quad \text{tr}(\mathbf{A}) = A_{ii}, \quad \mathbf{A} : \mathbf{B} = A_{ij}B_{ij}.$$

$$(\mathbf{AB})_{ij} = A_{ik}B_{kj}.$$

$$\mathbf{x} \cdot \mathbf{Ax} = x_i A_{ij} x_j.$$

$$\nabla \mathbf{A} \quad \text{for a vector field gives} \quad (\nabla \mathbf{A})_{ij} = \partial_j A_i.$$

$$\nabla \cdot \mathbf{T} \quad \text{for a second-order tensor can be written} \quad (\nabla \cdot \mathbf{T})_i = \partial_j T_{ij}.$$

1.5 Gradient, Divergence and Curl in Orthogonal Coordinates

The following formulas use physical components A_1, A_2, A_3 along unit vectors $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$.

General Orthogonal Differential Operators

$$\begin{aligned}\nabla f &= \sum_{i=1}^3 \hat{\mathbf{e}}_i \frac{1}{h_i} \frac{\partial f}{\partial q^i}. \\ \nabla \cdot \mathbf{A} &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q^1} (h_2 h_3 A_1) + \frac{\partial}{\partial q^2} (h_3 h_1 A_2) + \frac{\partial}{\partial q^3} (h_1 h_2 A_3) \right]. \\ \nabla \times \mathbf{A} &= \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\mathbf{e}}_1 & h_2 \hat{\mathbf{e}}_2 & h_3 \hat{\mathbf{e}}_3 \\ \partial_{q^1} & \partial_{q^2} & \partial_{q^3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}.\end{aligned}$$

1.6 Cylindrical Differential Operators

Cylindrical: (ρ, ϕ, z)

$$\begin{aligned}\nabla f &= \hat{\mathbf{e}}_\rho \frac{\partial f}{\partial \rho} + \hat{\phi} \frac{1}{\rho} \frac{\partial f}{\partial \phi} + \hat{\mathbf{k}} \frac{\partial f}{\partial z}. \\ \nabla \cdot \mathbf{A} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}. \\ \nabla \times \mathbf{A} &= \hat{\mathbf{e}}_\rho \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \hat{\phi} \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) + \hat{\mathbf{k}} \left(\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{1}{\rho} \frac{\partial A_\rho}{\partial \phi} \right).\end{aligned}$$

1.7 Spherical Differential Operators

Spherical: (r, θ, ϕ)

$$\begin{aligned}\nabla f &= \hat{\mathbf{e}}_r \frac{\partial f}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi}. \\ \nabla \cdot \mathbf{A} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}. \\ \nabla \times \mathbf{A} &= \hat{\mathbf{e}}_r \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \\ &\quad + \hat{\mathbf{e}}_\theta \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] + \hat{\mathbf{e}}_\phi \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right].\end{aligned}$$

Quick Drills and Practice: Coordinates and Tensor Operations

Level A: Coordinate conversion

1. Convert $(x, y, z) = (3, 4, 5)$ to cylindrical coordinates.
2. Convert $(x, y, z) = (1, 1, \sqrt{2})$ to spherical coordinates.
3. Convert $(\rho, \phi, z) = (2, \pi/3, 5)$ to Cartesian coordinates.
4. Convert $(r, \theta, \phi) = (4, \pi/2, \pi/6)$ to Cartesian coordinates.
5. Fill in: $\rho = \underline{\hspace{2cm}}, z = \underline{\hspace{2cm}}, r = \underline{\hspace{2cm}}.$
6. Fill in: $dV_{\text{cyl}} = \underline{\hspace{2cm}}.$
7. Fill in: $dV_{\text{sph}} = \underline{\hspace{2cm}}.$
8. Fill in the spherical scale factors.

Level B: Scale factors and metric matrices

1. Starting from $\mathbf{r} = \rho \cos \phi \hat{\mathbf{i}} + \rho \sin \phi \hat{\mathbf{j}} + z \hat{\mathbf{k}}$, derive h_ρ, h_ϕ, h_z .
2. Starting from $\mathbf{r} = r \sin \theta \cos \phi \hat{\mathbf{i}} + r \sin \theta \sin \phi \hat{\mathbf{j}} + r \cos \theta \hat{\mathbf{k}}$, derive h_r, h_θ, h_ϕ .
3. Write the metric matrix $[g_{ij}]$ for cylindrical coordinates.
4. Write the metric matrix $[g_{ij}]$ for spherical coordinates.
5. Find ds^2 for cylindrical coordinates and explain why the term $\rho^2 d\phi^2$ appears.
6. Find ds^2 for spherical coordinates and explain the term $r^2 \sin^2 \theta d\phi^2$.

Level C: Differential operators

1. Compute ∇f in cylindrical coordinates for $f = \rho^2 z \sin \phi$.
2. Compute ∇f in spherical coordinates for $f = r^2 \cos \theta$.
3. Compute $\nabla \cdot \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = \rho^2 \hat{\mathbf{e}}_\rho + \rho z \hat{\phi} + z^2 \hat{\mathbf{k}}$.
4. Compute $\nabla \cdot \mathbf{A}$ in spherical coordinates for $\mathbf{A} = r^2 \hat{\mathbf{e}}_r + r \sin \theta \hat{\mathbf{e}}_\theta + 0 \hat{\mathbf{e}}_\phi$.
5. Compute $\nabla \times \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_\rho + \rho^2 \hat{\phi} + 0 \hat{\mathbf{k}}$.
6. Compute $\nabla \times \mathbf{A}$ in spherical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_r + 0 \hat{\mathbf{e}}_\theta + r \sin \theta \hat{\mathbf{e}}_\phi$.

Level D: Tensor operations

1. Given $A_{ij} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 4 & 0 & 2 \end{pmatrix}$, compute $A_{ij}x_j$ for $\mathbf{x} = (1, 2, 3)$.
2. Compute $\text{tr}(A)$, $A : A$, and A^2 for the matrix above.
3. If $T_{ij} = x_i x_j$, compute $\partial_j T_{ij}$ in three-dimensional Cartesian coordinates.
4. For $\mathbf{A} = x^2 \hat{\mathbf{i}} + xy \hat{\mathbf{j}} + yz \hat{\mathbf{k}}$, compute $(\nabla \mathbf{A})_{ij}$.
5. Show explicitly that $\nabla \cdot \mathbf{A} = \text{tr}(\nabla \mathbf{A})$ for the same vector field.